

Harry Nguyen

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SUMMARY

CS @ Georgia State (May 2027, GPA 3.75) – Google Chrome SWE intern, primary engineer at TiMoto AI. Ships end-to-end production systems fast and works directly with users to define what to build. Experienced owning architecture decisions with no safety net across backend, cloud infra, and ML serving.

EDUCATION

Georgia State University

Expected May 2027

Bachelor of Science in Computer Science • **GPA: 3.75/4.0**

- Coursework: Distributed Systems, Operating Systems, Database Systems, Computer Networks, Data Structures
- Honors: Presidential Scholarship, Anita Andrew Fellowship, Roxie Alexander Memorial Award

EXPERIENCE

TiMoto AI | Software Engineer

Sep 2025 – Present

- Owned end-to-end delivery of production AI applications – collaborated with end-users to define, architect, and ship **distributed systems from scratch** on a 3-person engineering team
- Designed **vLLM inference engine with PagedAttention** – evaluated vLLM vs naive inference, selected for KV cache efficiency under concurrent load; deployed with continuous batching achieving **zero OOM failures** at **sub-50ms p99**
- Diagnosed distributed deadlock under concurrent gRPC calls – traced shared resource acquisition conflicts, redesigned call sequencing – restored **100% evaluation success rate** at **sub-50ms p99**
- Architected multi-AZ ECS Fargate with **Terraform IaC**, CloudWatch observability, and circuit breaker pattern – **99.9% uptime**, **44% cost reduction** (\$40–60/month); auto-rollback on health check failure
- Maintained distributed production systems across gRPC, PostgreSQL, and ML serving – on-call rotation, root-cause analysis, runbook documentation, post-mortem learnings applied to future design

Google | Software Engineer Intern – Chrome Browser

May 2025 – Aug 2025

- Shipped **C++ IPC transport layer** with Protocol Buffers to Chrome stable channel – **3B+ active users**, **sub-50ms p99**, **10K+ req/sec** – in 12-week internship; selected Protocol Buffers over custom serialization for schema evolution
- Identified settings navigation as p99 bottleneck at 1,200ms; designed **lock-free concurrent trie search** eliminating mutex contention – **96% latency reduction**, zero production regressions
- Architected event-driven TypeScript/React system with observer pattern – accelerated feature delivery **68%** across **25K+ lines** of Chromium at **95% test coverage**
- Delivered design documents and code reviews; changes adopted into production branch reviewed by senior Chrome engineers

Develop for Good | Software Engineer Intern

May 2024 – Aug 2024

- Collaborated with nonprofit program managers to define requirements and shipped a stateless BaaS on AWS – auto-scaling supporting **500+ concurrent users**, **90% deployment time reduction** via CI/CD (GitHub Actions)
- Diagnosed N+1 query bottleneck degrading response to 3s+ on large datasets; redesigned with PostgreSQL indexing achieving **sub-100ms** for 10,000+ records

PROJECTS

Pulumi – Contributor | Go, TypeScript, K8s, IaC, Multi-cloud, CLI

github.com/pulumi

- Submitted Go CLI features and bug fixes enabling multi-cloud (AWS/Azure/GCP) provisioning – under active review by core maintainers (24.4K+ star project)
- Analyzed **Raft/Paxos consensus** in distributed state synchronization layer – studied linearizability guarantees and correctness under concurrent operations and partial failures

SKILLS

Languages: C++, Python, Go, TypeScript, Java, JavaScript, SQL, Bash, Rust

Backend & Systems: gRPC, Protocol Buffers, Circuit breakers, Exactly-once semantics, Fault tolerance, Kafka, Event-driven architecture

ML & AI Infrastructure: vLLM, PagedAttention, Continuous Batching, LoRA/QLoRA, PyTorch, LangChain, LLM-as-a-judge evaluation

Cloud & Infrastructure: AWS (ECS Fargate, EKS, EC2, S3, RDS), Terraform, K8s, Docker, CI/CD, CloudWatch, Prometheus, Grafana

Frameworks & Databases: Django, FastAPI, Node.js, React, REST APIs, PostgreSQL, Redis, MongoDB

Build & Delivery: Rapid prototyping, end-to-end ownership, design docs, code review, post-mortem process

AI Dev Tools: Claude Code, GitHub Copilot, Codex, Cursor